

Artificial Intelligence and Machine Learning Models for Forecasting, Optimization, and Control in Smart Energy Systems

Grzegorz Dudek ^{1,2,3,*}  and Marcin Blachnik ^{4,*} 

¹ Faculty of Electrical Engineering, Czestochowa University of Technology, Al. Armii Krajowej 17, 42-201 Czestochowa, Poland

² Faculty of Mathematics and Computer Science, University of Łódź, ul. Banacha 22, 90-238 Łódź, Poland

³ CAMINO—Centre for Data Analysis, Modelling and Computational Sciences, University of Łódź, ul. Narutowicza 68, 90-136 Łódź, Poland

⁴ Department of Industrial Informatics, Silesian University of Technology, ul. Akademicka 2A, 44-100 Gliwice, Poland

* Correspondence: grzegorz.dudek@pcz.pl (G.D.); marcin.blachnik@polsl.pl (M.B.)

1. Introduction

The global energy sector is undergoing a profound and multidimensional transformation driven by decarbonization policies, increasing electrification, large-scale integration of renewable energy sources, and the growing digitalization of energy infrastructures. These changes have significantly increased the complexity of energy systems, which must now operate under high uncertainty, strong temporal variability, and tight economic, environmental, and reliability constraints. As a result, traditional model-based and rule-based approaches are often insufficient to fully capture the nonlinear dynamics and stochastic behavior inherent in modern energy systems.

In this context, artificial intelligence (AI) and machine learning (ML) have emerged as key enabling technologies for the next generation of smart energy systems. By leveraging large volumes of heterogeneous data—from sensors, supervisory control systems, weather services, and energy markets—AI-driven models offer powerful tools for forecasting, optimization, control, and decision support. Recent advances in deep learning, ensemble methods, reinforcement learning, and hybrid modeling have further expanded the applicability of data-driven approaches across the entire energy value chain, from generation and transmission to consumption and market operations.

Despite their growing adoption, the deployment of AI and ML methods in energy systems presents several challenges. Energy-related data are often incomplete, noisy, and nonstationary, while operational environments are subject to physical constraints, safety requirements, and regulatory frameworks. Moreover, many energy applications require models that are not only accurate but also robust, computationally efficient, and interpretable, particularly when used to support real-time operation or economically critical decisions. These challenges have motivated the development of tailored learning architectures, application-aware evaluation metrics, and hybrid approaches that combine data-driven learning with domain knowledge.

Against this backdrop, this Special Issue, entitled “Artificial Intelligence and Machine Learning Applications in Smart Energy Systems”, brings together nine contributions that address a broad spectrum of problems in contemporary energy systems using advanced AI and ML techniques. The papers cover a wide range of application domains, including renewable energy forecasting, oil and gas production prediction, nuclear integrated energy systems, hydropower operation, data center energy management, power system



Received: 29 December 2025

Accepted: 15 January 2026

Published: 2 February 2026

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protection, electrical load forecasting, and climate change adaptation. Methodologically, the contributions span deep neural networks, gradient-boosted decision trees, hybrid learning architectures, reinforcement learning, and ensemble-based approaches, reflecting the methodological diversity of the field.

A distinguishing feature of this Special Issue is its emphasis on practical relevance and real-world deployment. Several studies explicitly consider data scarcity, computational constraints, and economic performance metrics, while others rely on high-fidelity digital twins or long-term operational datasets. This focus underscores a broader shift in energy-related AI research—from proof-of-concept studies toward solutions that can be reliably integrated into operational energy systems. At the same time, the included works highlight the importance of aligning algorithmic innovation with system-level objectives such as efficiency, resilience, safety, and sustainability.

The aim of this extended Editorial is to synthesize the contributions of the Special Issue, place them within the broader international research landscape, and identify emerging trends and open challenges in the application of AI and ML to smart energy systems. Rather than providing isolated summaries, this Editorial seeks to highlight conceptual connections across the papers, revealing common methodological themes and complementary perspectives. By doing so, it aims to offer readers a structured overview of current advances and to stimulate further research at the intersection of artificial intelligence and sustainable energy systems.

The remainder of this Editorial is organized as follows: Section 2 presents a thematic summary of the nine contributions, grouped according to their primary methodological focus and application domain. Section 3 discusses cross-domain themes and conceptual integration, highlighting shared challenges and research directions that transcend individual applications. Finally, Section 4 concludes with reflections on future opportunities and the role of AI and ML in shaping the evolution of smart, resilient, and low-carbon energy systems.

2. Summary of the Contributions

2.1. *Learning-Based Forecasting for Energy Generation and Demand*

Accurate forecasting of energy generation and demand remains a foundational requirement for the reliable, efficient, and economically optimal operation of modern energy systems. The increasing penetration of renewable energy sources, the decentralization of generation assets, and the growing sensitivity of electricity markets to forecast errors have significantly raised the performance expectations placed on predictive models. Within this context, four contributions in this Special Issue address forecasting problems across diverse energy domains, employing advanced machine learning architectures while explicitly considering practical challenges such as nonlinearity, data scarcity, computational efficiency, and application-driven evaluation criteria.

Qiu et al. focus on oil well production forecasting, a task characterized by strong temporal dependencies, complex nonlinear relationships among variables, and imperfect data quality. The authors propose a hybrid GRU–KAN architecture that integrates gated recurrent units for temporal feature extraction with Kolmogorov–Arnold networks for nonlinear functional representation. A key novelty of this work lies in the explicit separation of temporal dynamics and nonlinear approximation, enabling the model to capture both sequential dependencies and intricate input–output mappings. The use of Particle Swarm Optimization to tune model parameters further enhances predictive performance while maintaining computational efficiency. By validating the approach on real oil well datasets and demonstrating superior accuracy over conventional neural and hybrid models, the

study illustrates the value of modular hybrid architectures in production forecasting for the oil and gas sector.

Short-term renewable energy forecasting is addressed by Kopyt et al., who conduct a systematic comparison of gradient-boosted decision tree models for wind power prediction. Unlike many studies that focus on a single boosting framework, this work evaluates multiple GBDT variants—including classical GBDT, XGBoost, LightGBM, and CatBoost—under identical experimental conditions. The novelty of this contribution lies not only in the breadth of the comparison but also in its emphasis on computational performance alongside forecasting accuracy. By explicitly analyzing training time and hyperparameter sensitivity, the authors provide insights that are directly relevant for operational deployment, where model retraining frequency and computational cost are critical factors. The results demonstrate that model choice involves trade-offs that extend beyond raw predictive accuracy, reinforcing the need for holistic evaluation frameworks in renewable energy forecasting.

Lee and Jeong address a different but increasingly important challenge in forecasting: data scarcity in small-scale photovoltaic systems. Their work introduces a forecasting framework that combines transfer learning with TSMixer architectures and dynamic time warping-based similarity assessment. The key contribution of this study lies in its ability to leverage data from multiple source domains to improve prediction accuracy in target systems with limited historical records. By quantifying similarity between time series and selectively transferring knowledge, the proposed approach mitigates one of the most common barriers to deploying data-driven models in distributed, small-scale energy systems. The demonstrated superiority of the approach over conventional LSTM and Transformer models highlights the importance of domain adaptation strategies in renewable energy forecasting.

Electrical load forecasting, a cornerstone of power system operation and market design, is examined by Beloev et al. using XGBoost models combined with carefully designed preprocessing and tuning procedures. While gradient boosting has been widely applied in load forecasting, the novelty of this work lies in the introduction of a financial-loss-based evaluation metric tailored to balancing energy market operations. By showing that models optimized using traditional error metrics do not necessarily minimize economic losses, the authors underscore a critical gap between statistical accuracy and operational effectiveness. This application-driven perspective provides an important complement to purely technical forecasting studies and demonstrates how machine learning models can be aligned more closely with market and system-level objectives.

Taken together, these four contributions reveal several important connections and shared themes. First, all studies move beyond generic forecasting pipelines by tailoring model architectures, optimization strategies, or evaluation metrics to the specific characteristics of their application domains. Second, they collectively illustrate a shift toward hybrid and ensemble-based thinking, whether through combining temporal and functional modeling (GRU-KAN), benchmarking multiple boosting frameworks, integrating transfer learning with temporal mixers, or coupling prediction accuracy with economic impact. Third, the papers emphasize that forecasting performance must be assessed not only in terms of statistical error but also with respect to computational feasibility, data availability, and real-world operational consequences.

From a broader perspective, this group of papers reflects the maturation of learning-based forecasting in energy systems. Rather than focusing solely on methodological novelty, the contributions demonstrate how machine learning models can be systematically adapted to address domain-specific constraints and objectives. In doing so, they provide valuable guidance for both researchers and practitioners seeking to deploy reliable forecasting solutions across heterogeneous energy applications.

2.2. Optimization and Control Using Reinforcement Learning

The second thematic group illustrates the growing role of reinforcement learning (RL) as a practical tool for optimization and control in modern energy systems. In contrast to classical optimization approaches that rely on explicit mathematical formulations (e.g., linear or mixed-integer programming) and require accurate system models, RL offers an alternative paradigm in which control policies are learned directly from interaction with an environment. This is particularly attractive in energy settings where dynamics are nonlinear, constraints are complex, and operational objectives must be balanced under uncertainty and changing market conditions. At the same time, applying RL in energy systems is nontrivial: the action space is often continuous, safety and feasibility constraints are strict, and training must be conducted in a way that avoids damaging real assets. The two contributions in this section address these challenges from complementary angles—economic dispatch in nuclear integrated energy systems and operational dispatch control in cascade hydropower—together demonstrating how careful reward design, realistic simulation environments, and systematic benchmarking can make RL viable for high-impact energy applications.

Arvanitidis and Alamaniotis consider the economic dispatch problem for nuclear integrated energy systems (NIESs) that incorporate Small Modular Reactors (SMRs) alongside other assets and grid interactions especially including renewable energy sources (RESs). Economic dispatch in such systems is complicated by the multi-sector nature of NIES, the variability in external conditions and market prices, and the need to optimize profitability while maintaining reliable baseload operation. The authors propose an off-policy RL framework augmented with an ensemble reward system. The novelty here is not simply the use of RL but the recognition that reward engineering becomes a central bottleneck in complex energy optimization problems: single-reward formulations may bias learning toward short-term profit at the expense of stability, feasibility, or long-term economic performance. By constructing an ensemble of reward components and integrating them into off-policy learning, the proposed approach aims to stabilize training and improve sample efficiency while capturing multiple operational priorities (e.g., economic performance, dispatch feasibility, and system competitiveness). The study demonstrates that this integrated reward strategy yields more accurate and efficient dispatch decisions than conventional approaches, highlighting RL's potential for profit-driven optimization in spot market environments. More broadly, the work advances the emerging view that reward shaping and multi-objective balancing are decisive elements in bringing RL from laboratory settings to operational energy markets.

Rot Weiss et al. address a fundamentally different but equally challenging problem: dispatch control for a cascade hydropower plant system using deep reinforcement learning within a digital twin environment. Cascade hydropower operation involves complex nonlinear flow dynamics, strong coupling among plants along a river, and operational decisions that must align with both demand schedules and market requirements. Traditionally, such control is performed by skilled dispatchers who rely on domain expertise and heuristics. A primary innovation of this paper is the creation and use of a high-fidelity, data-based digital twin environment of eight hydropower plants on the Drava river, which enables safe and realistic RL training. The authors evaluate multiple state-of-the-art RL algorithms (DDPG, TD3, SAC, and PPO), providing an informative comparison that highlights algorithmic trade-offs under realistic operational constraints. Another distinctive contribution is the explicit benchmarking against human dispatcher performance, positioning the RL agent not as an abstract optimizer but as a practical decision-support or automation candidate. The reported results—an absolute mean error comparable to human dispatchers—suggest

that well-trained RL agents can approach expert-level operational behavior in complex, coupled energy systems, especially when trained in realistic simulation settings.

Viewed together, these two contributions reveal several important conceptual connections. First, both studies emphasize that environment design and learning objectives are as important as the choice of RL algorithm itself. Arvanitidis and Alamaniotis focus on reward construction to encode economic and operational priorities, whereas Rot Weiss et al. focus on digital twin realism to ensure that learned policies reflect the true dynamics of a physical energy system. Second, both papers demonstrate that RL in energy systems should be evaluated not only in abstract terms (e.g., cumulative reward) but also against meaningful operational baselines: conventional dispatch methods in the NIES setting, and human dispatcher performance in the hydropower setting. Third, both studies implicitly address the critical challenge of deploying RL in safety- and reliability-critical infrastructures by shifting training to simulation-based environments and by shaping objectives to discourage unsafe or infeasible behavior.

Beyond their immediate application domains, the papers jointly contribute to a broader methodological narrative: RL is increasingly positioned as a flexible framework for energy optimization under uncertainty, particularly when traditional modeling is incomplete or overly restrictive. At the same time, these works underscore key research advancements for the wider community, including robust multi-objective reward design, constraint-aware RL, sim-to-real transfer and validation, and the integration of RL agents into human-in-the-loop operational workflows. By demonstrating RL's feasibility for both market-oriented dispatch (SMR-based NIES) and physically constrained flow control (cascade hydropower), the two studies provide complementary evidence that reinforcement learning can serve as a unifying paradigm for next-generation optimization and control across heterogeneous smart energy systems.

2.3. Energy Efficiency, Reliability, and Infrastructure Resilience

The third thematic group addresses a set of interrelated priorities that increasingly define modern energy systems: improving energy efficiency, strengthening infrastructure reliability, and enhancing resilience under operational stress, uncertainty, and climate-driven variability. While forecasting and control are essential for day-to-day operation, long-term system performance depends equally on the ability to monitor, model, and protect critical infrastructure, as well as to design decision-making processes that remain robust under evolving external conditions. The three contributions in this section span distinct physical contexts—data center cooling, superconducting fault current limiters, and climate adaptation through renewable integration—yet they converge on a common methodological agenda: using data-driven models to increase situational awareness, reduce risk, and support engineering decisions in complex cyber-physical systems.

Kula et al. focus on a highly practical and energy-intensive application domain: thermal management in data centers. Cooling systems account for a substantial fraction of data center electricity consumption, and even modest improvements in predictive control can translate into significant energy savings and reduced carbon footprint. The authors develop and evaluate machine learning models to forecast the temperature within a hot (or warm) corridor based on external weather conditions and internal operational variables such as server energy consumption and cooling system state. A notable contribution of this work is the careful treatment of the end-to-end modeling pipeline in an operational setting: the paper describes dataset construction for a functioning data center, which is often a major barrier to reproducible research in this domain. From a methodological standpoint, the study compares modern time series neural architectures (TiDE and TSMixer) with classical machine learning approaches (random forest, XGBoost) and statistical baselines

(AutoARIMA). The results demonstrate that TiDE achieves the lowest overall prediction error, while the analysis also reveals an important operational nuance: XGBoost tends to underestimate temperature at higher values, a bias that may be unacceptable in safety-critical thermal management. This combination of comparative benchmarking with an explicit discussion of risk-relevant error behavior constitutes a key novelty: rather than treating forecasting error as a purely statistical quantity, the authors interpret model behavior in relation to operational hazards and constraint violations. In doing so, the paper contributes not only a predictive model comparison but also a decision-oriented perspective on model selection for energy efficiency and reliability.

Hajdasz et al. address infrastructure resilience from a different angle: the reliability and durability of superconducting fault current limiters (SFCLs) in smart power systems. SFCLs are promising technologies for improving power network selectivity and reliability, yet their behavior is governed by nonlinear and dynamic physical processes that are difficult to model with conventional engineering equations alone, particularly the degradation of high-temperature superconducting (HTS) tapes due to cyclic transitions into the resistive state. The authors propose a concept of an engineering decision-support system (EDSS) that aims to predict degradation using macroscopically measurable quantities such as dissipated energy and the number of resistive transitions. Within the scope of a preliminary study, they evaluate multiple data-driven approaches, including Gaussian process regression (with different kernels), k-nearest neighbors regression, random forests, PCHIP interpolation, and polynomial approximation, under conditions of limited experimental data. The paper's novelty lies in demonstrating that even relatively simple regression techniques—especially Gaussian process regression—can provide useful degradation inference when the direct measurement of critical current is infeasible. Importantly, the authors explicitly frame their work as a step toward decision support rather than a purely predictive exercise, emphasizing uncertainty considerations and the need for expanded datasets and physics-informed constraints in future EDSS development. This contribution therefore advances the discourse on how AI can support reliability engineering in emerging grid technologies, where data may be scarce and the cost of failure is high.

Rojek et al. broaden the perspective to the system level by examining how machine learning can support next-generation climate change adaptation through increased renewable energy integration and improved efficiency. Their article synthesizes key application areas such as predictive maintenance, intelligent grid management, forecasting of supply and demand, and real-time optimization of renewable resources. Beyond cataloging technical methods, the authors emphasize challenges that frequently determine real-world impact but are often under-discussed in purely algorithmic studies: data availability and standardization, model transparency, and the need for interdisciplinary collaboration across engineering, policy, and regulation. A distinctive contribution of this review is its explicit linkage between methodological advances and strategic implementation conditions. By highlighting that successful climate adaptation requires not only predictive and control algorithms but also infrastructure readiness, structured datasets, and alignment with ethical and regulatory frameworks, the paper strengthens the bridge between AI research and actionable energy transition planning. In the context of this Special Issue, it provides a unifying macro-level rationale for why the other contributions—focused on forecasting, optimization, and component-level reliability—matter for broader societal goals.

Across these three papers, several conceptual connections emerge. First, all contributions position AI and ML as tools for decision support under operational constraints, rather than as purely academic forecasting or pattern recognition exercises. In data centers, prediction errors translate into energy waste or safety risks; in SFCLs, model uncertainty influences reliability assessment and component lifetime decisions; and in climate adap-

tation, algorithmic performance must be interpreted through the lens of implementation feasibility and governance. Second, the papers collectively illustrate the importance of model robustness under limited or imperfect information. Kula et al. confront real-world sensor and operational variability; Hajdasz et al. operate under small-sample experimental regimes; and Rojek et al. discuss data limitations and transparency as systemic barriers to deployment. Third, the papers collectively emphasize that evaluation criteria should be application-aware: beyond average prediction error, one must consider risk-sensitive bias (e.g., underestimation near safety limits), uncertainty quantification (critical for degradation prediction), and policy and infrastructure alignment (critical for climate adaptation).

Together, these contributions highlight a critical dimension of intelligent energy systems: resilience is built not only through better forecasts or smarter controllers but also through reliable infrastructure modeling, safety-aware decision-making, and an understanding of the broader socio-technical context in which algorithms are deployed. By combining operational case studies with component-level reliability modeling and system-level climate adaptation perspectives, this group of papers demonstrates how AI and ML can contribute to energy efficiency and resilience in complementary and mutually reinforcing ways.

3. Synthesis and Research Directions Across Forecasting, Control, and Resilience

Despite the diversity of application domains covered in this Special Issue, the nine contributions converge on a coherent set of cross-cutting themes that characterize the state of the art in AI- and ML-enabled smart energy systems. These themes reflect not only algorithmic progress but also a broader shift in how energy researchers frame predictive and decision-making problems, moving from purely accuracy-driven modeling toward deployable, risk-aware, and system-integrated intelligence.

A first unifying theme is the transition from forecasting as a standalone task to forecasting as a component embedded in operational decision loops. Across renewable generation, load prediction, industrial production, and thermal management, the studies in this Special Issue implicitly treat forecasts as inputs to downstream actions—dispatch, balancing, economic planning, safety control, or resource allocation. This aligns with the growing consensus that evaluation must incorporate deployment constraints and operational objectives rather than focusing solely on point-wise statistical errors. In practical settings, forecast errors translate into economic losses, constraint violations, or safety risks; therefore, application-aware metrics and decision-centric validation protocols become essential. The Special Issue illustrates this direction particularly clearly through the integration of forecasting accuracy with market-relevant performance criteria and risk-sensitive error behavior.

A second theme is methodological pluralism paired with engineering realism. Instead of promoting a single model class as universally superior, the contributions collectively demonstrate that modern energy problems often benefit from a portfolio of approaches: gradient-boosted trees for robust tabular forecasting; sequence models (e.g., GRU-like architectures) for extracting temporal dependencies; transfer learning for data-scarce assets; and reinforcement learning for sequential decision-making. This mirrors a broader trend in the ML community, where boosting remains a strong baseline for structured data and operational pipelines [1,2], while deep learning architectures offer flexibility in representation learning but require careful design and validation.

A third integrative motif is the increasing prominence of learning under constraints: data scarcity, nonstationarity, safety requirements, and physical feasibility. Several papers explicitly address limited historical data (e.g., small-scale PV or specialized industrial settings) through transfer learning, similarity measures, or careful feature construction.

Others operate under strict safety constraints (data center temperature risk, grid component degradation) where conservative bias and uncertainty awareness may matter more than average accuracy. These concerns resonate with a data-centric view of AI, emphasizing that data quality, provenance, and operational alignment frequently determine real-world impact more than incremental model sophistication [3]. In smart energy systems, where measurement and logging infrastructures vary widely, this perspective is particularly important for building reliable and maintainable AI solutions.

A fourth theme is the emergence of simulation-enabled learning and digital twins as a practical bridge between data-driven methods and physical infrastructures. Reinforcement learning contributions in this Special Issue rely on high-fidelity simulation to enable safe training, systematic evaluation, and comparison against operational baselines. This reliance on simulated interaction reflects established RL principles [4] and aligns with the broader engineering movement toward digital twins as virtual representations that support monitoring, prediction, and optimization across system lifecycles [5]. For energy applications, digital twins can serve not only as training environments but also as validation tools for stress-testing policies against rare events, abnormal operating regimes, and constraint boundaries that are difficult to observe in historical data.

A fifth theme concerns the integration of domain knowledge and physical structures with machine learning. While not all contributions explicitly adopt physics-based constraints, the direction toward hybrid modeling is evident: models are tailored to domain structures (e.g., cascade coupling, thermal behavior, degradation mechanisms) and evaluated with respect to physically meaningful behaviors. This broader trajectory is increasingly reflected in the literature on physics-informed learning, which seeks to combine data efficiency with physical consistency and improved generalization [6]. For smart energy systems—where conservation laws, operational constraints, and safety margins are non-negotiable—hybrid and physics-informed approaches represent a promising path to reducing brittle behavior under distribution shifts.

Finally, this Special Issue collectively highlights a practical research agenda for the next phase of AI in energy: (i) decision-centric evaluation that links predictive performance to economic and safety outcomes; (ii) robust learning under distribution shift and data scarcity; (iii) constraint-aware and safe reinforcement learning for mission-critical control; (iv) standardized, reusable digital twin environments for benchmarking; and (v) hybrid modeling that incorporates physical principles and uncertainty quantification. These directions align with recent surveys emphasizing reinforcement learning's expanding role in sustainable energy systems and the need for standardization and safety in sequential decision-making under uncertainty [7].

In summary, the contributions of this Special Issue support a consolidated view of smart energy intelligence: forecasting, optimization, and reliability are not separate research silos but complementary components of an integrated socio-technical system. Progress, therefore, depends on methods that are accurate and operationally meaningful, computationally feasible and robust, and innovative and aligned with engineering realities.

4. Conclusions

The nine contributions collected in this Special Issue demonstrate the breadth and maturity of contemporary AI and ML research in smart energy systems. From forecasting and optimization to infrastructure reliability and climate adaptation, the papers showcase how advanced learning methods can address pressing challenges in modern energy systems.

Looking ahead, future research will benefit from deeper integration of physical modeling and data-driven learning, paying increased attention to interpretability and trust, and closer alignment between technical innovation and policy objectives. We hope that

this Special Issue serves as both a reference and a catalyst for further advances in artificial intelligence and machine learning for sustainable energy systems.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflicts of interest.

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